

Infrared Image Super-Resolution: Systematic Review, and Future Trends

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Abstract—Image Super-Resolution (SR) is essential for a wide range of computer vision and image processing tasks. Investigating infrared (IR) image (or thermal images) super-resolution is a continuing concern within the development of deep learning. This survey aims to provide a comprehensive perspective of IR image super-resolution, including its applications, hardware imaging system dilemmas, and taxonomy of image processing methodologies. In addition, the datasets and evaluation metrics in IR image super-resolution tasks are also discussed. Furthermore, the deficiencies in current technologies and possible promising directions for the community to explore are highlighted. To cope with the rapid development in this field, we intend to regularly update the relevant excellent work at https://github.com/yongsongH/Infrared_Image_SR_Survey.

Index Terms—Image Super-resolution, Deep Learning, Convolutional Neural Networks (CNN), Generative Adversarial Nets (GAN)

1 INTRODUCTION

IMAGE super-resolution reconstruction (SR) is a classic low-level vision task in image processing [1]–[4]. The super-resolution reconstruction of IR images aims to recover high-resolution (HR) IR images from low-resolution (LR) IR images [1], [2]. Due to the existence of diverse degradation paths, the super-resolution reconstruction for IR images is still considered to be an ill-posed problem [2], [5], [6]. More details of the problem definition are shown in Eq. 1:

$$I_{LR} = \mathbb{D}(I_{HR}; \delta) \quad (1)$$

where $\mathbb{D}$ denotes a degradation function, I_{HR} is the high-resolution IR image, I_{LR} is the low-resolution IR image and δ is the parameters of the degradation process. In general, the degradation process includes the following factors: 1) downsampling, 2) noise, and 3) compression, as shown in Eq. 2. In different real-world settings, the importance of these factors will be different. Visualization for the IR image super-resolution degradation and reconstruction can be found in Fig.1.

$$\mathbb{D}(I_{HR}; \delta) = (I_{HR} \otimes k) \downarrow_d; [k, \downarrow_d] \subset \delta \quad (2)$$

where $I_{HR} \otimes k$ represents the convolution between a blur kernel k and the HR image I_{HR} . In the k , noise and compression are included. And, $\downarrow_d$ is a downsampling factor, e.g., $4\times$ and $8\times$. Briefly, the super-resolution reconstruction objective function of IR images can be described as Eq. 3:

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(I_{HR}, I_{SR}) + \lambda \Phi(\theta) \quad (3)$$

where $\mathcal{L}$ denotes the loss function, between the HR image I_{HR} and the SR image I_{SR} . $\Phi(\theta)$ and λ are the regularization term and punishment parameter, respectively. Initial works follow the IR images SR as general image SR, and simply adopt the visible images SR model (e.g., Convolutional Neural Networks (CNN)-based methods [7], Generative Adversarial Nets (GAN)-based methods [1], [8], [9], and transformer model [10]) to the new task.

However, it should be pointed out that the degradation process in infrared images is more complicated. The main reasons include the following: first, the blur kernel used in the reconstruction of visible images is relatively simple. It is typically Gaussian noise and JPEG compression [9], [11]–[14]. But, these blur kernels are not robust enough for IR images to represent complex real-world conditions. Because IR imaging systems are limited by optical physics and electronics. In other words, the noise generated is different. In addition, many IR cameras are deployed in the wild environment. It further increases the complexity of IR image degradation. On the other hand, the IR image presents simple color, insignificant gradients, and information overlap between high and low frequencies [11], [15]–[26]. These characteristics, which are different from visible images, can be considered to be unique patterns in IR images. Consequently, it is necessary to propose super-resolution

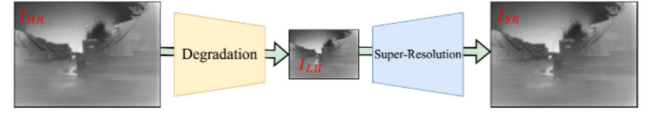


Fig. 1: Degradation and reconstruction in IR image super-resolution.

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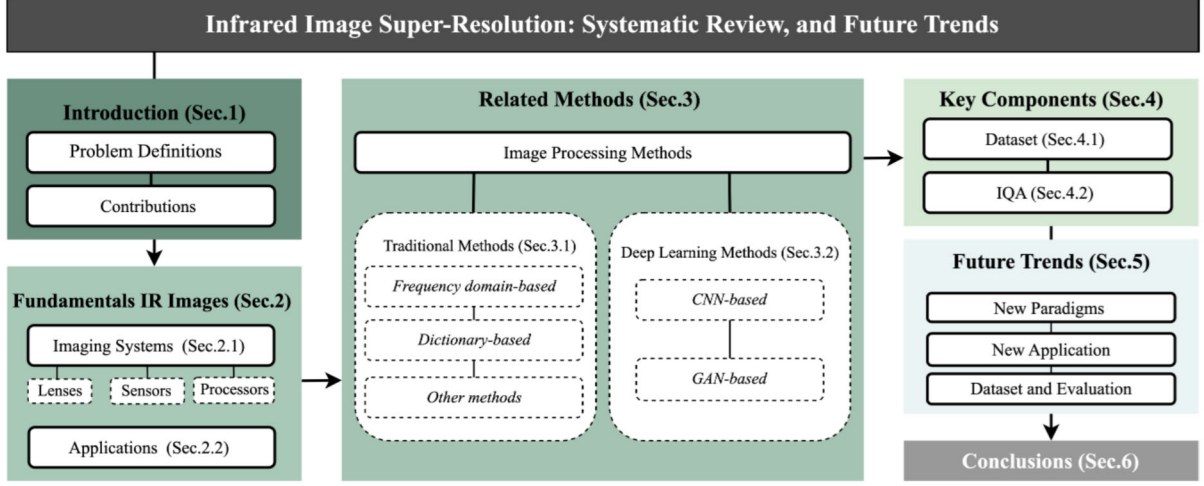


Fig. 2: Hierarchical and structural taxonomy of this survey.

reconstruction approaches appropriate for IR images. We will discuss more details in future sections.

In this paper, we systematically review applications, methods, and challenges in the field of IR image super-resolution. Unlike other reviews, our survey focuses on recent deep-learning algorithms and summarizes the specific patterns used in IR images for model representation. It will benefit the whole community to propose algorithms that are different from visible image reconstruction to improve the quality of IR images. To the best of our knowledge, this is also the first work that provides the most detailed explanation of why enhancing hardware systems to improve IR image quality is difficult. Unlike existing SR surveys [1]–[3], [27], [28], this survey uses the perspective of the IR image’s unique patterns to illustrate the challenges that distinguish it from visible images and also aims to review recent advances in SR techniques for IR images in the deep learning field. Meanwhile, we also summarized the application of IR images in various engineering fields. It will help the researchers to have a comprehensive insight into the entire field. The hierarchical and structural taxonomy of this survey is shown in Fig.2.

This survey’s contribution can be summarized as the following:

- 1) We systematically review the applications, methods, datasets, and evaluation metrics related to IR image super-resolution. In particular, we carefully investigated the application in IR image super-resolution using relevant methods since the beginning of the deep learning explosion. It includes the classification and summarization of these methods.

- 2) We also provide some unique pattern details in IR images that will benefit reconstruction, which are features making IR images super-resolution different from visible image super-resolution. It is common to see poor performance when directly transferring methods from visible images to IR images. It is promising to be solved by focusing more on the unique features of IR images.

- 3) We discuss trends, open issues, and challenges in IR image super-resolution. This section is dedicated to providing the community with new perspectives and new

directions to explore. We will continue to update our open-source repository with new work in this area, and hope it will be useful for future highlighting research. The repository link is https://github.com/yongsongH/Infrared_Image_SR_Survey.

The remainder of this survey is organized as follows: the imaging systems and IR image applications are introduced separately in Section 2; Then, Section 3 describes image processing methods for IR image super-resolution in the literature; In Section 4, we will present the key components in IR image super-resolution: datasets and image quality assessment (*i.e.*, IQA) metrics; IR image’s specific patterns, future trends, and open issues will be discussed in Section 5; finally, the work is summarized in Section 6.

2 FUNDAMENTALS OF INFRARED IMAGES

In this section, we will introduce the fundamentals of IR images, which are IR imaging systems and applications. For the imaging system, we will discuss the components in the system and how these items influence the IR image quality. Then, IR image applications in different engineering fields will be classified and then discussed.

2.1 Imaging Systems

In the classical IR imaging system, the following basic components are included: lenses, sensors, processors, and others [2], [80]. In Fig. 3, we show the details in a classic forward-looking infrared (FLIR) system. On the linear array of IR detectors, the objective produces an image. Each detector’s amplified output drives a corresponding light-emitting diode (LED). After reflection from the back of the scanning mirror, this LED array is transferred onto an image tube (vidicon). In turn, this powers a typical cathode ray tube display. The whole system converts a scanned infrared image into a digital representation that can be seen on a monitor [80]. At present, the most popular infrared imaging systems use incorporate solid-state detectors.

We can achieve the goal, which is improving the IR image quality, by enhancing the performance of imaging systems. According to current research, some key components

TABLE 1: IR image super-resolution applications.

Medical biochemical engineering	Vision tasks	Other engineering tasks	
Pharmaceutical industry [29], [30]	Image conversion [31]	Automated vehicle [32]–[34]	Food quality control [35], [36]
Medical science [37]–[40]	Multispectral matching [41], [42]	Remote sensing [43]–[47]	Agricultural management [48]–[54]
Cellular observations [55]–[57]	Targets detection [58]–[60]	Terrain models [45]	Water resource management [61]–[63]
Fluorescence microscopy [64]–[68]	Face recognition [69]–[73]	Land surface [46], [74], [75]	Star formation [76]–[79]

in the imaging system: **lenses, sensors, and processors**, are significant in improving IR image quality. However, we will face many challenges in improving the image quality directly from the imaging system. More details will be discussed in the next sections.

2.1.1 Difficulties w.r.t. Lenses

In general, we need more light to be captured by the imaging system to get better image quality [80], [81]. But more light means a larger lens, which brings two main challenges: First, the more sophisticated instrumentation required will increase the costs and difficulties in the entire industrial design [82], [83]. Secondly, larger optical lenses make the size of the detector also larger [81]. It makes the whole size of the imaging system becomes larger and the combination of both becomes more difficult [84]. In motion situations, such as fast-moving environments, less light will arrive at the machine, making the image quality poor [81], [85].

2.1.2 Difficulties w.r.t. Sensors

With the captured IR light, the sensor is a key component to translate the light-wave signal to quantified signal. This component also involves two following major difficulties in improving the imaging performance. Firstly, there is the inherent noise of the sensor [86]–[90]. It is a noise called fixed pattern noise (FPN), which exists in the differences between bias voltages in column readout circuits of uncooled infrared sensors resulting in strong stripe noise which changes slowly in time [87], [91]. If we address this problem from hardware, it is almost impossible to solve it completely due to the physical limits [89], [92], [93]. Secondly, the limitation of device size makes it difficult for the current technology to make advances in the short term [81], [88].

2.1.3 Difficulties w.r.t. Processors

For many IR imaging systems, the traditional microprocessors are still equipped. Therefore, it may be difficult to run the sophisticated SR algorithms at the stage of image capturing, because of the limited computing power itself [92], [94].

Improving the performance of the equipment can be promptly expressed in the image quality improvement. However, as mentioned above, we will face many difficulties with the key components used in IR camera imaging systems. It mainly includes the challenge of the fabrication process and physical limitations. For the fabrication process, we expect the camera to be more portable but often require larger lenses for better imaging. For processors, we need new technologies to improve computing and processing power, which requires more time and effort.

In summary, improving imaging quality from hardware will be costly in terms of industrial expense and effort, and the final performance improvement will be limited due to insurmountable physical obstacles. Based on the above concerns, we would resort to the subsequent image processing technology to improve the IR image quality.

2.2 Applications

IR image super-resolution reconstruction techniques have critical applications in various fields. In this section, we will show the applications in these key areas as shown in Tab. 1. In summary, they can be generally divided into three categories: medical biochemical engineering, vision tasks, and other engineering tasks.

2.2.1 Medical Biochemical Engineering

For medical biochemical engineering, there are two typical examples presented here: neurodegenerative diseases and COVID-19 diagnosis. Super-resolution reconstructed infrared images can be used in the diagnosis of neurodegenerative diseases [37]. As a non-invasive examination approach, it helps to visualize and provide evidence for consultation. The current global epidemic COVID-19 diagnosis also requires the help of IR images. Because CT images are costly for large-scale population screening, the use of infrared imaging can provide a replacement. The super-resolution reconstructed IR images are also used to extract time-series features, which are finally used as a reference for classification using machine learning methods [38], [39]. Moreover, high-resolution IR images are fundamental for the visualization of living cells [55], [95], protein structure analysis [96], [97], and nanoscale drug [98]. These studies can be further developed with the help of super-resolution reconstruction in IR images.

2.2.2 Vision Tasks

For vision tasks, super-resolution reconstruction of infrared images also plays an important role. We will present several representative tasks: multispectral matching, image conversion, and object detection. Multispectral matching [41], [42] is popularly used in disaster prevention and homeland security. Better resolution of infrared images means better matching results. It is also beneficial for the object detection and other tasks. By using IR images, it is able to identify objects unaffected by the weather, illumination, or the environment's color. And, image conversion can convert electro-optical (EO) images to IR images [31]. The image conversion is significant because we use IR images for recognition, but we often receive EO images. On the other hand, IR images are very critical for object detection in some environments.

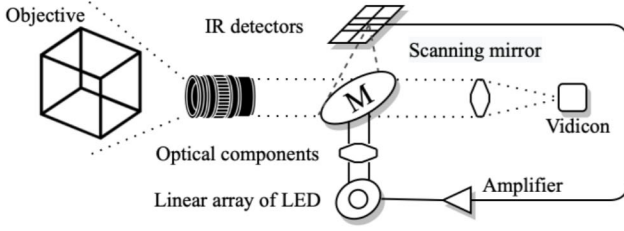


Fig. 3: Explanation of the operation mechanism in a FLIR system.

For example, penetrating IR images are the key information source after disasters such as earthquakes. With the help of HR images, we can locate trapped people in dark environments, such as mines, by using localization algorithms [59], [99], [100].

2.2.3 Other Engineering Tasks

For other engineering tasks, we can find the value of high-quality IR images in different disciplines. In the automated vehicle, infrared images in low-visibility conditions would be beneficial for assisted driving systems, but high-resolution thermal imaging sensors are typically expensive, which limits the general availability of such imaging systems [32]. Therefore, super-resolution techniques can be used to improve performance in this field [101], [102]. Further, super-resolution IR images have the potential to provide substantial improvements to interferometric observations of protoplanetary disks [103]. Moreover, estimates of relative dust optical depth and source brightness are also important [77]. Ultra-high-resolution IR images are also needed in the study of ongoing star formation [104], [105]. Applications in other domains can be found in more detail in Tab. 1.

3 RELATED METHODS

In this section, we will introduce IR image super-resolution methods, including traditional methods and deep learning-based methods. For the traditional algorithms, there will be three parts: frequency domain-based, dictionary-based, and other methods. Then, the deep learning methods will be presented.

Based on the improved IR imaging system at hand, researchers are also focused on the more efficient image processing methods that can be used to enhance IR image quality. For image processing methods, we will show the two types of approaches, including traditional methods and others related to deep learning. There are some representative works listed in Tab.2 to compare the performance.

3.1 Traditional Methods

In traditional image processing algorithms, the IR image super-resolution follows the direction of visible image super-resolution. Specifically, these approaches include **frequency domain-based, dictionary-based, and other methods**.

3.1.1 Frequency Domain-based

For the frequency domain-based approach, we will decompose the IR image into two domains: the air domain, and the frequency domain [126]. Especially for multiple complex frequency domain components, it is often not included by a simple FFT. For example, there is a significant difference between high-frequency and low-frequency information in IR images. *Choi et al.* propose a new frequency-domain processing method for this problem: firstly, the edge pixels are distinguished, then the high-frequency information is enhanced independently, and finally, a high-resolution image is available [127].

$$e(\hat{k}, l) = \left(1 - e^{-\frac{(\lambda_1 + \lambda_2)}{\sigma_{\text{sum}}}}\right) \left(1 - e^{-\frac{(\lambda_1 + \varepsilon)/(\lambda_2 + \varepsilon)}{\sigma_{\text{ratio}}}}\right) \quad (4)$$

As shown in Eq. 4, the researchers introduced prior knowledge skillfully. Two eigenvalues are denoted by λ_1 and λ_2 , where $\lambda_1 \geq \lambda_2 \geq 0$, which are recalculated from the structure tensor at pixel $(\hat{k}, l)$, σ_{sum} and σ_{ratio} indicate established parameters. In order to avoid division by zero, a tiny constant named ε is created. It is important to mention that we will need a carefully engineered prior with expert knowledge to ensure the performance, which is difficult to achieve with the current relatively weak physical modeling [2]. On the other hand, there are also some frequency domain-based methods focusing on the reconstruction time issue and aiming to accelerate [128]. Because we will fuse 4 images to get better image resolution. In addition, since the edges of the original image are bold, it will be difficult to match the edges during registration.

3.1.2 Dictionary-based

Considering the good interpretability, the dictionary-based reconstruction methods have gained a lot of popularity among researchers in IR images. The formalization of the dictionary-based reconstruction method can be found in this work [129]. In summary, the dictionary-based approach focuses on patches that build a bridge between I_{HR} and I_{LR} . These patches include features such as 1) correspondence, 2) the number of patches, and 3) inter-patch relationships. The correspondence relationship and the number of patches are used to ensure that the mapping between the I_{LR} and the I_{HR} is complete. And the patterns of the low-resolution image can be preserved in the high-resolution image by the inter-patch relationship. The proposed methods improve the image quality from these perspectives, and more details are shown in Tab. 3.

We will briefly describe the representative dictionary-based works: **First, dividing the groups**. On the one hand, it has been observed that the similarity between image groups impacts the reconstruction, that is, similarity and compatibility as previously discussed. This approach considers group similarity and proposes positional constraints. The position between groups is used as a feature to constrain the reconstruction of the image [119]. In addition, multiple information (multi-view) is introduced to reconstruct images using more groups to represent instead of isolated groups [120], [121]. Researchers use multi-scale groups to compile dictionaries, which can provide rich scale information and diverse structural information [121]. In

TABLE 2: Performance comparison [in terms of PSNR (dB) and SSIM] over selected test dataset for 4 or 3 upscaling factors. Some of these works use visible images [106] as training datasets, such as 7 and 9. The test datasets for these representative methods depend on the task, check the references for more details.

No.	Methods	Year	Ref. $\times 4$ (PSNR/SSIM)	Keywords
1	Wu, Wenhao, et al. [107]	2022	31.69/0.7877	Meta-learning; Lightweight network
2	Huang, Yongsong, et al. [108]	2021	33.13/0.8282	Transfer learning; Small sample
3	Liu, Qing-Ming, et al. [109]	2021	32.23/0.8720	Attention mechanism; GAN
4	Prajapati, Kalpesh, et al. [14]	2021	34.90/0.9134	Attention mechanism; Channel splitting
5	Marivani, Iman, et al. [110]	2020	35.19/0.9888	Sparse coding; Image fusion
6	Marivani, Iman, et al. [26]	2020	34.49/0.9853	Multimodal image; Residual learning
7	Yao, Tingting, et al. [111]	2020	34.54/0.8807	Dictionary learning; CNN
8	Song, Pingfan, et al. [112]	2020	36.36/0.9796	Dictionary learning; Sparse representations
9	Batchuluun, Ganbayar, et al. [113]	2020	22.99/0.9760	Image deblurring; GAN
10	Rivadeneira, Rafael E, et al. [114]	2019	37.85/ $\emptyset$	Thermal images; CNN
11	Suryanarayana, Gunnam, et al. [19]	2019	31.40/0.9513	Discrete wavelet transform; CNN
12	Marivani, Iman, et al. [115]	2019	33.19/ $\emptyset$	Sparse coding; CNN
13	He, Zewei, et al. [116]	2018	36.02/0.9230	Multiple receptive fields; CNN
14	Sun, Chao, et al. [117]	2018	32.71/ $\emptyset$	Zoom mechanism; Transfer learning
15	Han, Tae Young, et al. [25]	2018	39.69/0.9582 ($\times 3$)	Frequency components; CNN
16	Zhao, Yao, et al. [118]	2016	30.69/0.9031 ($\times 3$)	Compressed sensing; Dictionary learning;

TABLE 3: Dictionary-based reconstruction methods.

Category	Method	Key idea
Patches	Cheng-Zhi, Deng, et al. (2014) [119]	Considering the similarity between groups, the locality-constrained group is proposed to guide the reconstruction by using the positions between groups as features.
	Yang, Xiaomin, et al. (2015) [120]	Introduce multi-view information and multiscale information to build patches.
	Yang, Xiaomin, et al. (2017) [121]	
Dictionary	Yang, Xiaomin, et al. (2016) [122]	Using more dictionaries.
	Yao, Tingting, et al. (2020) [111]	It is proposed to capture more IR image features with neural networks and then use this information to build dictionaries.
Correspondence Relationship	Zou, Erbo, et al. (2018) [123]	To better represent the pattern of groups, this work improves performance by learning the correspondence several times.
Extra Information	Song, Pingfan, et al. (2019) [112]	Other image modalities, such as RGB images, are introduced to enhance edge and structure primitives information.
	Chen, Zuming, et al. (2018) [124]	These works propose to pay more attention to the gradient information in IR images.
	Wang, Yan, et al. (2021) [125]	

specific tasks, such as pedestrian recognition, if the person as a target is segmented into different blocks will hurt the recognition results. In this situation, multi-scale blocks can keep the person retained in the complete block while further constructing the dictionary [123]. **Second, the relationship between the total number of dictionaries and reconstruction.** The researchers dropped the question that the single dictionary created from IR images has limited the development of dictionary-based methods or not. This limitation is expressed in two directions: fewer dictionary pairs and less information about the IR images. It is proposed that a single dictionary cannot represent all the structural information. And, multiple dictionaries should be used to support the reconstruction [122].

$$\begin{aligned}
\mathbf{x}^h &= \Psi_c^h \mathbf{z} + \Psi^h \mathbf{u}, \\
\mathbf{x}^l &= \Psi_c^l \mathbf{z} + \Psi^l \mathbf{u}, \\
\mathbf{y} &= \Phi_c \mathbf{z} + \Phi \mathbf{v}.
\end{aligned} \tag{5}$$

Moreover, it is also the first time that extra dictionaries are constructed using visible image information obtained from the same scene in a dictionary-based approach to help the reconstruction. **Third, the information, from other types of images, can also be used to support the dictionary construction.** Because the image information in the IR image itself is insufficient due to the poor imaging environment. Then, people started to focus on introducing extra information. This algorithm is developed to enhance the resolution of the target LR image with the aid of another guidance HR image modality [112]. More details are shown in Eq. 5.

TABLE 4: IR image super-resolution via other traditional algorithms.

Category	Method
Iterative	A. Fernandez, et al. (2017) [130]
	Ahmadi, Samim, et al. (2020) [131]
Sparse Representations	Wang, Peng, et al. (2021) [132]
Sparsity	Liu, Xingguo, et al. (2019) [22]
	Gngr, Alper, et al. (2019) [20]
	Liu, Xingguo, et al. (2018) [133]
	Chen, Shaojun, et al. (2019) [134]
Projection	Zhang, X. F., et al. (2019) [21]
	Liu, Jinsong, et al. (2016) [135]
Regularization	Yu, Hui, et al. (2013) [136]
	Hui, Yu, et al. (2014) [137]
	Wang, Yan, et al. (2022) [15]

They express the LR image patch $\mathbf{x}^l \in \mathbb{R}^M$ and HR image patch $\mathbf{x}^h \in \mathbb{R}^N$ of the same image modality, and the guidance HR patch of another different image modality $\mathbf{y} \in \mathbb{R}^N$. The definitions of the symbols can be found in Abbreviations and Notations. There are also works that noticed that gradient information of IR images would be beneficial for reconstruction [124], [125].

3.1.3 Other Traditional Algorithms

There are other traditional algorithms used for IR image super-resolution listed here, as shown in Tab.4. Some methods aim to transfer the super-resolution methods in visible images to IR images. **1) Iterative** methods are used in some biometric application situations, such as iris recognition [130], and laser line scanning thermography [131]. However, such methods are not fine-tuned according to the special patterns in IR image. **2) Sparse representations** are used in land remote sensing where the IR term with IR information is calculated by computing the normalized difference target index (NDTI). There are also swarm optimization algorithms applied to minimize the terms to get the land cover target mapping (LCTM) [15]. **3) In sparsity-based methods**, researchers pay more attention to the specific patterns in IR images. In such methods, it is still common to process information in the frequency domain. However, if we only consider the frequency domain, the contribution from the space domain information may be ignored. For example, the similarity between neighboring points is considered, different from the general frequency domain-based approach, where the image gradient represented by sparsity is mentioned [20], [22]. In addition, there are new directions, *i.e.*, using higher-order image derivatives to receive the image prior to sparsity [133], [134].

S. C. Park *et al.* described the details about the **projection method** in his work carefully [2]. Such methods require appropriate priors to guide the projection, and researchers considered the phase characteristics in images to improve the algorithm [21]. It is also proposed to introduce illumination sensitivity prior to construct iterations [135].

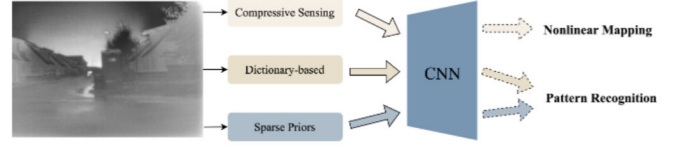


Fig. 4: CNNs contribute to other traditional methods used to learn nonlinear mappings and pattern recognition.

TABLE 5: CNN transfers on traditional methods.

Category	Method	Highlight
Compressive Sensing	Zhang, Xudong, et al. (2018) [23]	Nonlinear Mapping
Dictionary-based	Yao, Tingting, et al. (2020) [111]	Pattern Recognition
Sparse Priors	Marivani, Iman, et al. (2020) [110]	

On the other hand, **regularization** methods also play a key role. Yu Hui *et al.* [137] believed that the regularization term can also suppress the unnecessary low-frequency components and enhance the image quality. A two-step fully-variable sparse reconstruction algorithm is designed by Wang *et al.* [15] The fully-variable regularization term is added to the classical compressed perceptual model.

The weaknesses in such methods are also easily observed in experiments: Dijk *et al.* mention that regularization methods influence the performance in detection tasks [138]. Moreover, it is supposed that the model is not considered for complex nonlinear degradation factors. The result is that algorithms have limited fitness for real-world applications [136].

3.2 Deep Learning Methods

With the development of deep learning, neural networks have shown remarkable performance in image super-resolution with powerful fitting ability [7], [8], [139]–[142]. However, few previously published studies have been able to draw on any systematic research into IR image-specific patterns. These unique patterns include gradient information, high or low-frequency information, extra information, etc [21], [40], [108], [143], [144]. In this section, we will also focus on the differences between IR images and visible images in the deep learning field of super-resolution. In summary, the challenges of IR image super-resolution in deep learning include specific patterns, difficulty in representing patterns, and poor image quality. More details will be shown in the next section.

3.2.1 CNN-based

There are two major trends for CNN-based models in IR image super-resolution: **1)** First, researchers introduced CNNs as contributors to improving traditional algorithms (see Fig. 4, Tab.5). **2)** Further, it became popular to use CNNs to build end-to-end models used in IR image super-resolution.

CNN & Traditional algorithms: As mentioned in Sec .3.1, traditional methods require enough prior knowledge to achieve image super-resolution. However, weak mathematical analysis limits the proposed algorithms. CNN can learn nonlinear mappings without indication because of its

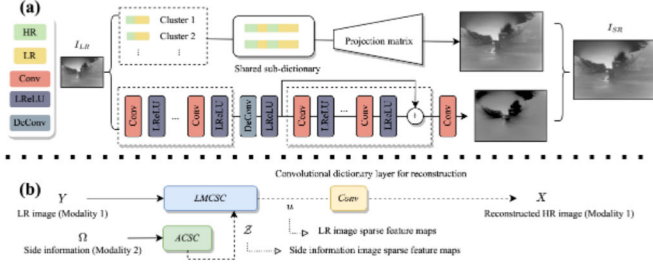


Fig. 5: (a) shows the CNN used as a feature extractor in a dictionary-based approach; (b) is an illustration for the CNN used to represent prior.

powerful fitting ability [7], [8], [139]–[142]. It allows people to free themselves from the task to seek for priors. Initially, neural networks were not used to reconstruct IR images directly. Rather, it was carefully used to help traditional algorithms improve performance through **nonlinear mapping and pattern recognition**.

1) *Nonlinear mapping*: In infrared camera systems, there will be little high-frequency information and low-frequency information in the captured infrared image compared to the visible image due to the poor imaging environment (see Sec.2.1). It becomes a challenge to better represent these two kinds of information in IR images super-resolution. Then, the researchers proposed that CNN can be used to represent the nonlinear mapping of low and high frequency information in the latent space. In *Zhang's work* [23], the SR image is reconstructed using compressed sensing first. Then the SR image and HR image are reduced to receive high-frequency noise information. Finally, the information is fed into CNN to learn nonlinear mapping.

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N \|\tilde{R}(\Theta, i) - R(\Theta, i)\|^2 \quad (6)$$

Eq.6 denotes the loss function to learn the trainable parameters Θ in CNN. Corresponding to i th training image, $\tilde{R}(\Theta, i)$ represents the estimated residual image produced by CNN, while $R(\Theta, i)$ represents the true residual image used for training [23]. Experiments show that this approach using CNNs to represent nonlinear mappings for reconstructing IR images can enhance detailed information.

2) *Pattern recognition*: On the other hand, *Yao, Tingting, et al.* [111] proposed to use CNN as a feature extractor in the dictionary-based approach. This method also focuses on the problem that the detailed information in IR images is important, and CNN can build the dictionary with more details added to the representation. Novel interpretable operators are proposed to construct the basic module as the sparse representation for prior extraction, another module to represent the edge information of visible images, and finally fused in a residual network using skip connection [110].

More details are shown in Fig.5. For (b), X and Y denote the LR image, and reconstructed SR image, respectively. The convolutional sparse codes Z of the guidance HR image Ω are similar to u by means of the ℓ_1 -norm according to the literature [110]. Furthermore, residual networks with

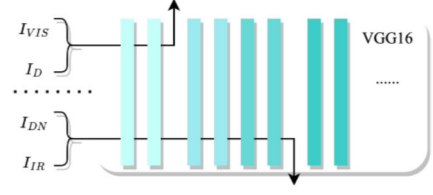


Fig. 6: I_{DN} denotes the denoised output, while I_D is the output from the author's proposed network [17].

multiple branches are used to reconstruct high and low-frequency information, respectively [17].

Beyond introducing CNNs, these existing methods started to introduce visible images to guide the reconstruction of IR images purposefully, which represents an entirely new direction. Further, attention has been focused on image characteristics, specifically the difference between high and low-frequency information in IR images. Details will be described in the next sections.

CNN & End-to-end models: In this section, the end-to-end models used in IR image super-resolution will be presented. The high-frequency and low-frequency information in IR images is responsible for the image outline and edges, respectively. Moreover, the IR image has less edge information compared to the visible image because of the imaging system (see Sec.2.1). **First, the researchers propose that the independent extraction for high-frequency and low-frequency information in IR images is achieved by CNN.** In [145], *Zou et al.* used residual networks to build a model similar to U-Net. The multi-receptive field module in this work is supposed to help represent the features of high and low-frequency information. Similar work is [144], and the residual network is also used. However, the information distillation approach is used in the backbone network, which was first proposed in [108] to benefit IR image super-resolution. This Information distillation-based model is also considered to be helpful for different information extraction. In addition, *Kwasniewska et al.* used a wide receptive field residual network constructed by dense connection [146]. According to the experimental results, this work confirmed that wide receptive fields are effectively used for low-contrast images. *Prajapati, Kalpesh, et al.* suggest that the common residual design allows for retaining too many redundant features in the network. A new module for integration with multiple attention mechanisms is proposed in their work. According to the experimental results, this novel attention mechanism module will help to enhance the high-frequency information representation [14]. Finally, not only the residual network is used, but also the importance of edges is observed [11]. The method that enhances high-frequency details and contacts information-tail modules to improve details is the highlight. More details are shown in Fig.7. And F_{nlmD} is the output of the previous (*i.e.* D) recursive block

The approaches mentioned above show that the two variables in the latent space (low-frequency and high-frequency information) are treated independently. Then novel methods are used to represent the nonlinear mapping between these two independent variables respectively. In

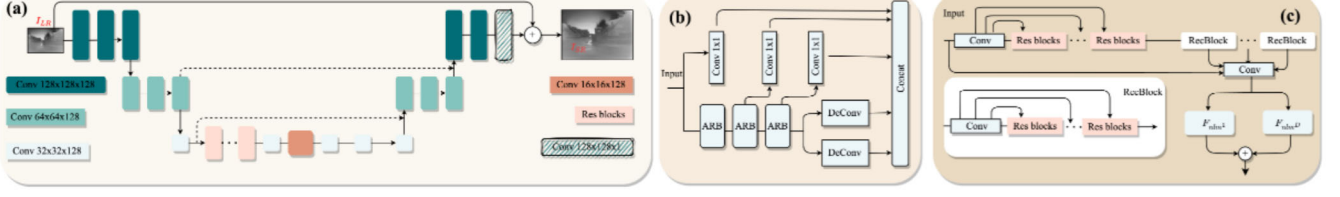


Fig. 7: In (a), the researchers used a network similar to U-Net to extract high and low-frequency information independently [145]. For the same purpose, (b) demonstrates that the information distillation method can also be used [146]. Finally, the module in (c) shows that wide receptive fields are beneficial for low-contrast images [11].

TABLE 6: Ablation studies for introducing visible image information. From 1 to 4 stars, indicates that the subjective evaluation is improving.

Residual Network	✓	✓	✓	
Visible Images (Training Dataset)		✓	✓	✓
Novelty Loss			✓	✓
Novelty Network				✓
Results	*	**	***	****

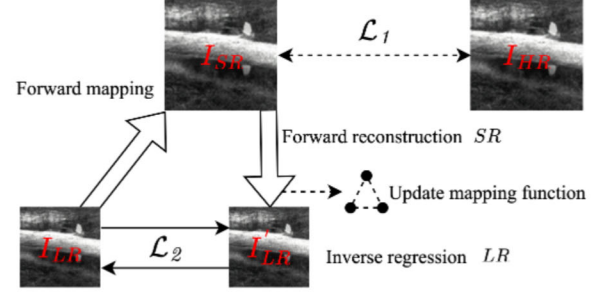


Fig. 8: Regression network structure.

the next section, we will describe another category of methods that introduce extra information: visible image information (see Tab.6).

For introducing visible image information, the researchers focused on the following components: dataset, network structure, and loss function. It is natural to associate using visible images to train neural networks because they are cheap and easy to use, with rich detailed information. However, it is difficult to fit the data distribution for IR images with the model trained on the visible image dataset due to the domain transfer challenge [108], [114]. In other words, the SR images have poor quality. Then, adaptations to the loss function are proposed. As we all know, the loss function is important for optimizing neural networks. *Patel, Heena M., et al.* proposed that a combined loss function, with two terms for L1 and SSIM loss, could be used [147]. According to the ablation experiments, this combined loss function would be beneficial for their proposed improved dense blocks. On the other hand, it also inspired that appropriate loss functions would be useful for introducing information from visible images In [17], the authors propose that the perceptual similarity between IR images I_{IR} and paired visible images I_{VIS} can be captured by CNNs. And there are features present in the middle layer of the neural network and they can be used as loss functions (see Fig.6).

Next, we will introduce novel network structures for using visible image patterns. *Zou, Yan, et al.* proposed a dual-path residual network that directly fuses the features from visible and IR images in the channel [148]. Further, *Oz, Navot, et al.* consider that dimensionality compression will influence the performance in feature fusion. Because the correlation between adjacent pixels is ignored. Then, the dimensional change network for increasing channels was proposed [149].

Finally, we will also present models related to super-resolution in multi-vision tasks. In many vision tasks [150], [151], such as segmentation and fusion, super-resolution

is applied to help other subjects as a pre-processing task. In particular, the regression network proposed in [151] is dedicated to prevent the irrelevant function mapping space on the reconstructed images by using forward generation and backward regression. This double mapping constraint is used for IR image super-resolution, and then the output that has better image quality is used in the fusion task. More details can be found in Fig.8.

The loss function used in this work is: x_i and y_i respectively represent the LR image I_{LR} and output SR images I_{SR} . $\mathcal{L}_1(F(x_i), y_i)$ and $\mathcal{L}_2(D(y_i), x_i)$ describe the loss functions of forward regression and inverse regression tasks, respectively.

$$\mathcal{L} = \sum_{i=1}^N \mathcal{L}_1(F(x_i), y_i) + \lambda \mathcal{L}_2(D(y_i), x_i) \quad (7)$$

Moreover, this work mentions that intensity distribution and gradient information can represent thermal radiation and structural information, separately.

In summary, the approaches in IR images and CNNs have first tried to combine the pattern recognition capabilities of CNNs into traditional reconstruction methods, such as dictionary reconstruction and sparse coding. Then started to reconstruct IR images directly using end-to-end models. Many module-based improvement methods were proposed, such as multi-scale information extraction. However, the high and low-frequency information in IR images have a significant gap. To use this information, the model structure has been improved further, for example, by presenting the information separately in different modules or using more complex modules. Compared to IR images, visible images have information with rich details. For this reason, researchers have started to try to use visible images to help reconstruct IR images. It includes designing new network

structures and using strategies such as transfer learning. The purpose is to help neural networks can introduce more patterns and information from visible images.

3.2.2 GAN-based

After GAN [152] was proposed, research based on generative models is emerging in the super-resolution field. SRGAN [8], ESRGAN [142], and various other types of GAN models [153] moved the field forward together through model improvement and mathematical analysis (WGAN [154]) modifications. The same attention has been focused on the application and research of GAN models in the IR image super-resolution.

Initially, GAN models using normal images have been directly used in IR image reconstruction. Researchers [155] used SRGAN straightly to reconstruct IR images and received reconstructed images that were quite acceptable. However, the blurred edges and unclear details are still a challenge. [156] used a modified DCGAN to reconstruct IR images, but the experimental results were not compared between the same category GAN methods. For this reason, it is difficult to describe the actual effectiveness. On the other hand, the work from [157] compared the existing SR algorithms cross-sectionally on IR images. The experimental results show that the SRFBN [158] model has the best universalization ability. But, for the GAN model, there are always unpleasant artifacts due to the model collapse. All these works illustrate the domain transfer difficulties possibly suffered by algorithms using visible images that are used in the IR images super-resolution task. Then, people started to consider the possibility of designing GAN models specifically for IR images. These studies are divided into the following categories: **module improvement**, **introduce extra information**.

1) *Module improvement*: It is natural first to consider that the improvement method is to explore new modules to improve performance. *Rivadeneira et al.* [159] uses the CycleGAN structure and employs ResNet as the generator's encoder. The self-attentive module is applied in the encoder and a new loss function is also proposed (see Eq. 8). According to the discussion in this work, the Sobel edge detector can capture the contour consistency between the input image and the cyclic-generated image by calculating the mean squared difference between the images.

$$\mathcal{L}_{\text{Sobel}} = \frac{1}{N} \sum_i \|\text{Sobel}(G_{H2L}(G_{L2H}(I_L))) - \text{Sobel}(I_L)\| \quad (8)$$

Another GAN model based on the attention mechanism has also been proposed, which differs in the loss function and discriminator used [109]. As shown in Fig. 9. Liu, Qing-Ming, et al. use a module based on the channel attention mechanism in the generator at their work. Moreover, they also replaced the original discriminator with a new one designed by the gradient penalty approach from WGAN. It will benefit network convergence. For the loss function, Wasserstein distance is used to evaluate the target image I_{HR} and the reconstructed image $\hat{I}$ (see Eq.9).

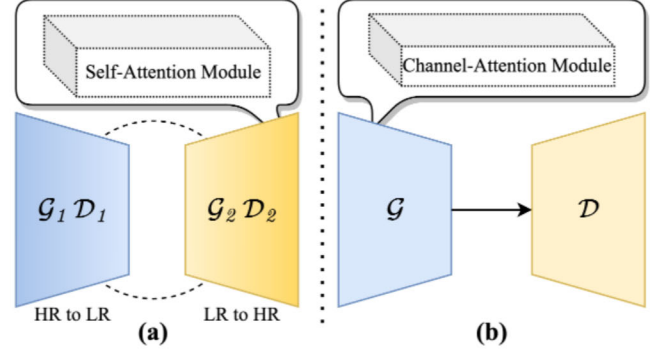


Fig. 9: Attentional mechanisms and GAN-based models for IR image super-resolution.

$$\mathcal{L}_{\text{WGAN}} = E[D_\theta(I_{HR}) - D_\theta(G_\theta(I_{LR}))] +, \quad (9)$$

$$\lambda E \left[\left\| \nabla_I D(\hat{I})_2 - 1 \right\|^2 \right]$$

where the second term on the right side of the equal sign is the gradient penalty term. Furthermore, WGAN is used to optimize the training process is also mentioned in this work [143].

2) *Introduce extra information*: This approach is focused on the issue that IR images have fewer patterns and further proposes to use visible images to guide the reconstruction. The main options include the following: **hybrid** and **split**. For the hybrid model, the features from visible and IR images are not purposely distinguished but mixed directly in the module. In [18], the information is initially extracted by residual blocks and then fused using 1x1 convolution. It is called multimodal, visual-thermal fusion model. Experiments show that visible images are beneficial for improving high-frequency details in IR images.

For the split model, *Huang et al.* [108] proposed using transfer learning to help introduce the patterns in visible images to reconstruct IR images. In this approach, PSRGAN, two components are proposed: the main path, which is responsible for extracting the features from visible images, and the branch path, which is used to represent the patterns from IR images. Finally, the IR image is reconstructed by a transfer learning strategy that maps both of them to a common feature space (see Multi-stage Transfer Learning Strategy 1). Such an approach can better capture and utilize the features in the model and show better results in the experiments. The input image data $x \in p_{data}$ and Z denote the output fake sample by generator G . Θ represents the parameters of G . And $\mathcal{L}$ is a loss function. E represents the cross-entropy.

Finally, there are also approaches that focus on the connection between SR methods and other tasks, such as denoising, and then models that combine cross-tasks are proposed. *Batchuluun, Ganbayar, et al.* [160] proposed methods that can be used for super-resolution and detection, where the super-resolution component focuses on both denoising and SR. In this work, the number of modules in the generator has been increased. In [161], researchers proposed a GAN-based framework for joint cross-modal

Algorithm 1 Multi-stage Transfer Learning Strategy

Input: $\lambda, p_{data}(x)$
Output: PSRGAN parameters Θ

- 1: **Init:** Initialize parameters Θ
- 2: **while** PSRGAN not convergent **do**
- 3: **for** Stage *I* not convergent **do**

$$\nabla_{\Theta} \mathcal{L}(G_1(Z_1; \Theta_1)),$$

$$+ \lambda \cdot \nabla_{\Theta} d_{\mathcal{KL}}(G_1(Z_1; \Theta_1), p_{data}(x)) \quad (10)$$
- 4: **return** $G_1(Z_1; \Theta_1)$
- 5: **for** Stage *II* not convergent **do**

$$\nabla_{\Theta} \mathcal{L}(G_2(Z_2; \Theta_2)),$$

$$+ \lambda \cdot \nabla_{\Theta} d_{\mathcal{KL}}(G_2(Z_2; \Theta_2) | (G_1), p_{data}(x)) \quad (11)$$
- 6: **return** $G_2(Z_2; \Theta_2)$
- 7: **for** Stage *III* not convergent **do**

$$\nabla_{\Theta} \mathcal{L}(G_3(Z_3; \Theta_3)),$$

$$+ \lambda \cdot \nabla_{\Theta} d_{\mathcal{KL}}(G_3(Z_3; \Theta_3) | (G_1, G_2), p_{data}(x)) \quad (12)$$
- 8: **return** $G_3(Z_3; \Theta_3)$
- 9: **return** $\Theta = (\Theta_1, \Theta_2, \Theta_3)$

and super-resolution aerial image vehicle detection. The first sub-network of this approach uses the GAN architecture to generate two different SR images by cross-domain transforms.

In summary, there is no doubt that the GAN model has great advantages in the field of generation, leading to a fairly good performance in SR algorithms. The results achieved in the visible field have attracted the application to the IR image super-resolution. However, according to the experimental results, the direct unmodified introduction of the normal image super-resolution reconstruction method encounters the difficulty of domain shift [155]–[157]. The edges in the reconstructed images are not clear. Therefore, new models have been proposed by considering the characteristics of IR images. These models are divided into improved modules [14], [109], [159], introduce extra information [18], [162], [163], and multitask [22], [113], [160], [161]. These algorithms have achieved some success, but missing standard benchmarks and datasets that are uniform leads to an unfavorable cross-sectional comparison between different types of methods. In the next section, we present available datasets and evaluation metrics in the IR image super-resolution field.

4 DATASETS & IQA

We will present the datasets and image quality assessment (IQA) metrics applied in the IR image super-resolution, respectively.

4.1 Datasets

For the super-resolution study, the datasets included: the training dataset and test dataset. Considering the training dataset, we hope that the samples are large enough to cover the current natural and realistic situations. It will be beneficial to the models' generalization. For the test dataset,

we would like to have standard datasets corresponding to different tasks. For example, in the face reconstruction task, the face dataset is important. But for iris reconstruction, the iris dataset would be more useful.

Datasets for IR images and video are available in other research topics [28], but fewer are focused on super-resolution. We will present the following representative work: *Weiss et al.* [164] present a publicly distributable dataset for the assessment of fusion, super-resolution, local contrast enhancement, dynamic range compression, and performance of image-based non-uniformity correction (NUC) algorithms. This dataset records images for different target background settings, camera or object movements, and temperature contrasts. Furthermore, *Alonso-Fernandez et al.* present 1,872 NIR iris image datasets [165]. *Almasi & Debeir* [42], on the other hand, presented a benchmark ULB17-VT dataset containing thermal images and their corresponding visual images. *Du, Juan, et al.* [166] also presented their dataset. For small sample reconstruction, *Huang, Yongsong, et al.* proposed the IR100 dataset [108]. *Huang, Yongsong, et al.* [143] further proposed a universal dataset including 1000 samples. The details are shown in Tab.7. Although there exists plenty of work on IR image super-resolution, it is limited in releasing the dataset.

For the test dataset, it is very common to select a small sample from the training dataset. These images are used as reference samples to evaluate the algorithm performance. On the other hand, IR image super-resolution is to be used in fused images sometimes. Considering this situation, there are also some algorithms [108], [143] that use fused images as reference, such as result-A [167] and result-C [168]. Different types of IR image samples will be useful to evaluate the reconstruction algorithm's robustness.

There are still some weaknesses in these datasets, which include the following aspects: these data are collected without indicating the relevant equipment parameters, which could influence the dataset selection in real situations. In addition, these datasets have different sample sizes. It is not conducive to compare them with each other. Furthermore, some datasets have no matching high-resolution visible image datasets, which causes difficulties in introducing the patterns from visible images. For datasets, we can make more efforts to complete the standard. On the other hand, it is also important to evaluate the reconstructed image quality. Then researchers also started to focus on proposing evaluation metrics applicable to infrared images which are also called IQA.

4.2 IQA

In the field of image reconstruction, how to evaluate reconstructed images is also a popular topic. We usually have three types of image evaluation metrics: **reference metrics**, **non-reference metrics**, and **human evaluation**.

4.2.1 Reference metrics

For reference metrics, the most commonly used to evaluate the algorithms' performance are peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). First, PSNR uses the maximum pixel value (denoted as L) and the mean square error (MSE) between images to estimate the reconstructed image quality. As shown in Eq. 13:

TABLE 7: Available dataset in IR image super-resolution fields. From left to right, the columns depict the name and the reference, pixel resolution of images, and a brief description, respectively

Name	Classes	pxl	Description
NATO SET-140 [164]	Universal	320x256 640x480 384x288x2	A publicly distributable dataset for assessment of fusion, super-resolution, local contrast enhancement, dynamic range compression, and image-based NUC algorithm performance.
CASIA Interval v3 [130]	Iris images	280x320	This database has 2,655 NIR images of 280320 pixels from 249 contributors captured in 2 sessions with a close-up iris camera, totaling 396 different eyes (the number of images per contributor and per session is not constant).
ULB17-VT [18]	Universal	240 x 320	Thermal and RGB images were manually extracted and annotated with a total of 570 pairs. The framework is divided into 512 training and validation samples and 58 test samples.
IR-COLOR2000 [166]	Universal	128x128	The training datasets (IR-COLOR2000), which contain 2000 pairs of images. The images in the datasets main contain the infrared images and corresponding RGB images.
IR100 [108]	Universal	640x480	They selected 100 samples from FIR Sequence Pedestrian Dataset as the images for small sample reconstruction.
CVC-09-1K [143]	Universal	640x480	The samples in this dataset come from 1000 randomly selected images from the FIR Sequence Pedestrian Dataset.
New dataset [114]	Universal	640x512	This dataset was acquired using a TAU2 thermal camera with a 13mm lens (45 HFOV) in a resolution of 640x512, with a depth of 8 bits and save it in PNG format
CDN-MRF dataset [116]	Universal	640x480	They build up a HR infrared image dataset covering a wide range of scenarios (e.g., vehicle, machine, pedestrian and building), and further present a DNN model to learn mapping relationship between LR and HR infrared images.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\frac{1}{N} \sum_{i=1}^N (I(i) - \hat{I}(i))^2} \right) \quad (13)$$

where L equals to 255 in general cases using 8-bit representations, I denotes the high resolution image while $\hat{I}$ represents the reconstructed image. Since PSNR is only related to pixel-level MSE and only cares about the differences between corresponding pixels without concerns about visual perception, it tends to lead to poor performance in reconstruction quality in real scenes, while we are usually more concerned with human perception. But in order to compare with other work, we always need to calculate this metric. Then, the metric closer to the human subjective evaluation was proposed: SSIM [169].

SSIM is dedicated to measuring structural similarity between images, based on independent comparisons with brightness $\mathcal{C}_l(I, \hat{I})$, contrast $\mathcal{C}_c(I, \hat{I})$ and structural $\mathcal{C}_s(I, \hat{I})$ aspects (see Eq.14).

$$\text{SSIM}(I, \hat{I}) = [\mathcal{C}_l(I, \hat{I})]^\alpha [\mathcal{C}_c(I, \hat{I})]^\beta [\mathcal{C}_s(I, \hat{I})]^\gamma \quad (14)$$

where α, β, γ are control parameters for adjusting the relative importance. More calculation details can be found in [169]. In short, SSIM evaluates the reconstruction quality from the perspective of human visual system [170]. Therefore, it better meets the requirements for perceptual evaluation [171], [172] and has been widely used in IR image super-resolution. In summary, the assessment of reference metrics always requires ground-truth images for comparison. In the real world, there is no ground-truth image. Then, no reference metrics are proposed.

4.2.2 No reference metrics

We will briefly describe the representative work of no reference metrics. The Natural Image Quality Evaluator (NIQE) [173] is a statistical collection of "quality-aware" features built on a simple but successful Natural Scene Statistics

(NSS) model in the spatial domain. The researchers also evaluated the perceptual image patch similarity (LPIPS) directly through the trained deep network based on the difference in depth features. According to the experimental results, it would be fairer to use the features extracted by the CNN to evaluate the image quality [174].

4.2.3 Human evaluation

Human subjective evaluation is the best indicator. In general, we need to evaluate and then score the reconstructed images. However, in order to exclude errors due to visual fatigue, we need more participants and more time to remove biased data. In other words, the time and resource cost for this evaluation method would be the most expensive.

4.2.4 Assessment of IR image super-resolution

For IR images, which suffer from missing rich details and colors. It may be difficult to indicate the real image quality by directly using the usual visible image metrics as evaluation.

In [175], the researchers propose a non-reference metric: Q_{SR} . It is actually combined previous works [176] and this metric can be used not only to help the algorithm converge but also to evaluate the image quality. As shown in Eq. 15, where λ_1 and λ_2 are hyperparameters. According to the literature [175], this metric takes the balance between the no-reference and reference metrics into consideration and provides a better assessment on the image quality.

$$Q_{SR} = \lambda_1 \frac{1}{\log_{10}(S_{BRISQUE})} + \lambda_2 S_{PSNR} \quad (15)$$

Further, the researchers proposed the phase in the IR image as the feature that can be used as an assessment metric to evaluate the image quality [21]. In [21], the characteristics in the frequency domain phase spectrum of IR images are considered to be different from the visible images. Such features, phase spectrum error (PSE), make it promising as an evaluation metric to be used specifically in IR images.

$$\text{PSE} = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N [P_f(i, j) - P_I(i, j)] \quad (16)$$

In Eq. 16, P denotes the phase spectrum, the M and N denote the number of rows and columns of the image, respectively.

Existing studies have focused on features in IR images that differ from visible images, such as phase similarity [21]. Quality assessment methods that differentiate from visible images are critical to advancing the field. However, researchers expect to observe the connection between IR and visible images, and proposed metrics that incorporate both will be attractive.

5 FUTURE TRENDS

In this work, we systematically review the excellent previous work on IR image super-resolution and make a comparison with the summary. Thus, we will introduce some attractive directions in this field and ideas that will further improve the model performance in this section.

5.1 New models & new paradigms

Good model design can improve the reconstructed image quality by better nonlinear mapping to represent complex patterns in high-dimensional space. For the new paradigm, we expect that the correlation between low and high frequency information in IR images will be studied to explain the correlation between them. We will present more details below.

Model design. With the booming in deep learning, more and more generative models are attracting attention in the community. Transformer [10], [177]–[179] and diffusion models [180]–[183] are representative works among them. We expect that the classical low-level vision task, IR image super-resolution, will benefit from these works.

For the transformer model, its excellent ability to capture long-range dependencies has been proven in almost all computer vision tasks [184]–[189]. There is no doubt that applying transformers in IR images will be valuable to try, after solving some difficulties. It is well known that the data-hungry problem is one of the difficulties plaguing transformers. In IR image super-resolution, massive datasets are challenging. Because expensive optical equipment is needed for capturing images that meet the specified requirements in appropriate outdoor conditions, as well as multiple scenes with increased time and energy requirements for the researcher. The modification to the transformer to make it available in small sample reconstructed scenes would be promising to try. Moreover, the large parameters in the transformer will make it strenuous to deploy in edge devices. Lightweight models are more valuable in the application to IR image super-resolution.

For the diffusion model, we expect its good interpretable performance to bring the IR image super-resolution to a new stage. As mentioned before, generative models, such as GAN, have been successful in IR images. However, the risk of model collapse still appears [108], [142], [143]. Although the gradient penalty is introduced, the generated image quality can still be further improved. If we can introduce

the diffusion model used in the IR image super-resolution, the mode collapse problem can be solved with the help of its good mathematical analysis. Remarkably, a skillful modification to make it capable of fitting the data distribution on IR images is necessary.

New paradigms. The unique pattern information in IR images is the coupling relationship between high and low-frequency information. It is a valuable problem to explore how to treat the frequency information in IR images. In previous work, they were treated independently or as independent variables in a high-dimensional space. It would be attractive to use a new paradigm to fit the gap between these two types of patterns. In particular, complex information in the latent space after representation may be useful to improve the detailed information in the reconstructed images. Moreover, interpretable works in deep learning incorporating traditional methods are also expected. Deep learning methods, which are often considered “black boxes”, may further clarify the details of nonlinear mapping in IR image super-resolution based on traditional methods with mathematical analysis. There is no doubt that it will be inspiring work in this field.

5.2 New application

With the development in the field of image super-resolution, blind super-resolution is the future-oriented direction [190]–[193]. Although many models or algorithms have been proposed for image super-resolution, the performance in the real world is not ideal. The main problem is that these algorithms simply summarize the degradation as bicubic downsampling and ignore the blurred kernels that exist in the real world, such as noise and compression [9], [13], [194]–[196]. Thus, studies on real-world super-resolution of visible images began to attract the attention of researchers. In IR images, to make the algorithms applicable in the real world, we can pay more attention to the degradation models of IR imaging and design the corresponding reconstruction algorithms. All these works will help the algorithms to face the complex degradation environment.

5.3 Dataset and evaluation metrics

On the one hand, datasets, the cornerstone for algorithm training, are fundamental to this task. As mentioned before, there are many works on IR image super-resolution that have not released datasets, for various reasons. Considering this situation, we need to establish benchmarks for this task. A robust enough dataset will be needed, including optical device details, image sizes, and other capture information. It will be beneficial to compare the performance between different types of algorithms. If a matching dataset with visible images, captured under the same environmental conditions, could also be included, there is no doubt that it would help to validate the effectiveness of introducing visible image patterns. Moreover, available datasets are also provided for thermal infrared image challenges as well. For example, the Thermal Image Super-Resolution (TISR) challenge organized in the Perception Beyond the Visible Spectrum (PBVS) is one of the promising events. This challenge provides the training dataset and the test dataset with 951 and 50 samples, respectively [197], [198].

On the other hand, researchers also need reliable evaluation metrics to evaluate the algorithm's performance. We can design specialized evaluation metrics for IR images to determine the reconstructed image quality, whether it is a reference metric or a non-reference metric. Although infrared images have unique pattern information compared to visible images, it would be pioneering work if new evaluation metrics could be used in image evaluation cross-domain.

6 CONCLUSION

In this paper, we gave a survey about the work related to the IR image super-resolution in the last decade. First, we discuss the fundamental role that the IR image super-resolution plays in the engineering field and other tasks. Then, we systematically summarize the factors that limit the imaging quality in IR imaging systems for the first time. It also explains why redesigning hardware systems to improve the IR image quality leads to high costs. Further, the image processing methods related to IR image super-resolution are classified and summarized. These works include traditional algorithms and deep learning-based methods. Moreover, we present the dataset and IQA which are components essential for the IR image super-resolution task. Finally, we point out some existing challenges for the future. We hope this survey will not only provide a better understanding of IR image super-resolution but also contribute to future development in this field and the real-world application with related algorithms.

Abbreviations.

Abbreviation	Full name
SR	super-resolution
IR	infrared
HR	high-resolution
LR	low-resolution
JPEG	joint photographic experts group
NUC	non-uniformity correction
IQA	image quality assessment
FLIR	forward-looking infrared
LED	light-emitting diode
CNN	convolutional neural network
GAN	generative adversarial net
WGAN	wasserstein GAN
SRFBN	feedback network for image SR
SRCNN	SR CNN
ResNet	residual network
DCGAN	deep convolution GAN
PSRGAN	progressive SR GAN
CycleGAN	cycle-in-cycle GAN
SRGAN	SR GAN
EO	electro-optical
FFT	fast Fourier transform
FPN	fixed pattern noise
NDTI	normalized difference target index
LCTM	land cover target mapping
PSNR	peak signal-to-noise ratio
SSIM	structural similarity index
MSE	mean square error
NIQE	natural image quality evaluator
NSS	natural scene statistics
LPIPS	perceptual image patch similarity

Notations.

Notation	Description
I_{LR}, X	low-resolution image
I_{HR}, I	high-resolution image
$\hat{I}, Y$	reconstructed image
$\mathbb{D}$	degradation function
$\otimes$	convolution operation
k	blur kernel
$\downarrow_d$	downsampling factor
$\mathcal{L}$	loss function
Φ	regularization term
λ	punishment parameter
θ, Θ	parameters of model
$\mathbb{R}$	image patch
$[\Psi_c^l, \Psi^l], [\Phi_c, \Phi]$	dictionary pairs
$u, v, \mathcal{Z}$	sparse codes
M, N	number of dimensions
i	i th training image
$\tilde{R}$	estimated residual image
R	true residual image
$\mathcal{L}_1$	loss functions of forward regression
$\mathcal{L}_2$	loss functions of inverse regression
$\mathcal{L}_{\text{Sobel}}$	loss functions of Sobel edge detector
G	generator
D	discriminator
I_L	generated low-resolution images
$\mathcal{L}_{\text{WGAN}}$	loss functions of WGAN
E	cross-entropy
p_{data}	input image data distribution
Z	distribution of output fake sample by generator
$d_{\mathcal{KL}}$	$\mathcal{KL}$ distance
∇	gradient operator
α, β, γ	control parameters
Ω, u	ℓ_1 -norm
$\mathcal{C}_l$	brightness similarity
$\mathcal{C}_c$	contrast similarity
$\mathcal{C}_s$	structural similarity
P	phase spectrum
$\lambda_1 \ \& \ \lambda_2$	eigenvalues
$\sigma_{\text{sum}} \ \& \ \sigma_{\text{ratio}}$	pre-defined parameters
ε	constant
e	edges
$(\hat{k}, l)$	pixel

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